

WEEK 3 INTERNSHIP PROJECT

TEST PLAN AND TEST CASE DEVELOPMENT

Internal Communication Platform

System Testing & Quality Assurance

Document Detail	Value
Project	Internal Communication Platform
Task	Week 3 — Test Planning and Case Development
Prepared By	Junior Systems Analyst
Document Version	2.0
Date	7 September 2026
Status	Test Plan for Testable System

Confidentiality: Academic / Internship Project — Synthetic test data only.

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1. Introduction and Background

The Internal Communication Platform is a proposed information system for a mid-sized company that needs a reliable and secure replacement for fragmented internal communication channels. Week 1 established the business and user requirements, while Week 2 translated those requirements into a proposed system architecture. Week 3 assumes that the system has now been developed to a testable stage.

This document defines how the system will be verified before release. It is written from the perspective of a Junior Systems Analyst supporting quality assurance and provides a repeatable basis for functional, non-functional, security, performance, and user acceptance testing.

1.1 Quality Goals

- Correctness — approved business functions operate as specified.
- Security — users can access only the data and actions authorized for their roles.
- Reliability — communication data remains consistent during normal and abnormal conditions.
- Performance — common actions remain responsive under expected concurrent usage.
- Usability — employees can complete common tasks without unnecessary complexity.
- Traceability — every major requirement has one or more verification activities.

2. Test Objectives

- Verify that implemented functions meet the approved Week 1 requirements.
- Validate end-to-end message, channel, announcement, notification, search and file workflows.
- Verify administrator functions and role-based authorization boundaries.
- Detect security weaknesses involving authentication, sessions, input handling and uploads.
- Measure response time, concurrency, throughput and error behavior under representative load.
- Confirm fixes through retesting and protect existing behavior through regression testing.
- Produce objective evidence supporting a release recommendation and UAT.

3. Scope and Assumptions

3.1 In Scope

- Authentication and session management
- Direct and channel messaging
- Message edit/delete
- Channels and membership
- Search
- Mentions and notifications
- Announcements
- File upload/download validation
- Administrator user lifecycle
- Role-based authorization
- Audit logging
- Performance and concurrency

- Recovery behavior
- Responsive browser usability
- Regression testing

3.2 Out of Scope

- Production penetration testing by an external security provider
- Certification of third-party infrastructure
- Legal/privacy certification
- Testing of unrelated enterprise applications
- Full disaster-recovery certification

3.3 Assumptions

- A stable staging build is available.
- Approved requirements and Week 2 design are the baseline.
- Synthetic test data is used.
- External identity, storage and notification services are available or safely mocked.
- Test results are recorded against a specific build/version.

4. Test Strategy and Methodology

A risk-based testing strategy will be used. Critical business and security controls receive priority, while representative lower-risk workflows are covered through functional and regression suites.

Testing Type	Approach
Functional	Verify individual features against expected business behavior.
Integration	Verify data exchange among identity, application services, database, storage and notification components.
System / End-to-End	Verify complete user journeys across the integrated application.
Regression	Rerun impacted and core tests after changes or defect fixes.
Performance	Measure response time, throughput, concurrency, resource utilization and errors.
Security	Verify authentication, authorization, sessions, input validation, upload controls and auditability.
Usability	Observe representative users completing common tasks across supported browsers.
UAT	Business representatives confirm that the system supports operational expectations.

5. Testing Levels and Types

- Unit tests are assumed to be completed by developers before system QA entry.
- Integration tests verify interfaces between services and external dependencies.
- System tests verify the integrated application in a production-like staging environment.

- Regression tests protect previously verified functionality after changes.
- Smoke testing is performed after every major deployment to confirm the build is testable.
- Exploratory testing may be used to investigate unexpected behavior around formal cases.

6. Roles and Responsibilities

Role	Responsibilities
Junior Systems Analyst / QA Tester	Prepare data, execute cases, record evidence, raise defects, retest fixes, maintain traceability.
Developer	Investigate and fix defects, provide build notes, support technical diagnosis.
System Administrator / DevOps	Maintain staging environment, logs, monitoring, deployment and access.
Security Reviewer	Review security findings, validate high-risk controls and remediation.
Product Owner / Business Representative	Clarify business rules, perform UAT, approve accepted deviations.
Project Lead	Coordinate schedule, review metrics, approve release recommendation.

7. Testing Environment

Component	Configuration / Requirement
Client devices	Windows 11 desktop/laptop; representative mobile browser
Browsers	Current supported Chrome and Microsoft Edge
Application	Staging deployment of Internal Communication Platform
Backend	Node.js/TypeScript services behind API gateway/load balancing
Database	Isolated PostgreSQL test database
Cache	Redis test instance
Object storage	S3-compatible test bucket with non-production files
Identity	Isolated SSO/OIDC test provider or approved test accounts
Network	HTTPS-enabled corporate-like network
Monitoring	Application logs, metrics, error monitoring and audit logs
Performance tooling	Approved load-testing tool with synthetic users

8. Test Data Requirements

- Standard employee, manager, administrator, inactive and restricted accounts.
- Users belonging to different departments and project channels.
- Representative direct messages, channel messages, mentions and announcements.
- Unique search keywords and date-filtered messages.

- Supported files, unsupported extensions, oversized files and safe security-test samples.
- Boundary-length messages and search strings.
- Synthetic dataset for at least 100 concurrent active users.
- Expected audit events for privileged actions.
- All credentials and personal data must be synthetic; no production data is permitted.

9. Entry and Exit Criteria

9.1 Entry Criteria

- Build is deployed successfully and smoke-tested.
- Environment, database, storage and identity dependencies are available.
- Test accounts and data are prepared.
- Approved test cases have been reviewed.
- Known environment blockers are resolved or formally accepted.
- Logging and monitoring are enabled.

9.2 Exit Criteria

- All Critical and High-priority cases have been executed.
- No open Critical defect remains.
- High defects are fixed or formally accepted by authorized stakeholders.
- Required security tests pass.
- Performance results meet targets or approved deviations are documented.
- Regression testing for fixes is complete.
- Test summary and release recommendation are reviewed.

10. Test Schedule

Phase	Planned Time	Activities	Output
Preparation	Day 1	Environment, accounts, data, case review	Ready-to-test confirmation
Functional/System	Days 2–3	Core workflows and integrations	Functional results + defects
Security	Day 4	Authentication, authorization, sessions, input and files	Security findings
Performance	Day 5	Concurrency, response time, throughput	Performance report
Regression/UAT	Days 6–7	Retesting, regression and business validation	UAT results
Closure	Day 8	Metrics, defect review and release recommendation	Final test summary

11. Detailed Test Cases

Execution rule: record the actual result, evidence and status for every case. A test passes only when all stated expected results are achieved and no blocking defect is introduced.

TC-001 — Successful employee authentication

Field	Details
Test Type	Functional
Priority	Critical
Preconditions	Active standard employee account; login service available.
Test Data	Valid synthetic employee credentials.

Execution Steps

1. Open the application login page.
2. Enter the active employee username.
3. Enter the correct password or complete the approved SSO flow.
4. Submit the login request.
5. Observe the landing page and available navigation.

Expected Result: User is authenticated, redirected to the correct home page, and sees only permissions assigned to the account.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-002 — Invalid credentials and account protection

Field	Details
Test Type	Security
Priority	High
Preconditions	Test account exists and security controls are enabled.
Test Data	Incorrect password; invalid username.

Execution Steps

1. Open login page.
2. Enter a valid username with an incorrect password.
3. Submit the request.
4. Repeat within the configured threshold using the test account.
5. Attempt login with an invalid username.

Expected Result: Access is denied; response does not disclose whether an account exists; configured rate limiting/lockout behavior is applied and attempts are logged where required.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-003 — Direct message send and persistence

Field	Details
Test Type	Functional
Priority	Critical
Preconditions	Two active employees exist and can message each other.
Test Data	Unique message text.

Execution Steps

1. Log in as Employee A.
2. Open Employee B direct conversation.
3. Enter the unique message.
4. Send it.
5. Refresh the page.
6. Log in as Employee B and open the conversation.

Expected Result: Exactly one message is stored, displays correct sender/time, persists after refresh, and is visible only to authorized participants.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-004 — Create and join project channel

Field	Details
Test Type	Functional
Priority	High
Preconditions	Manager account has channel-management permission.
Test Data	Unique channel name, description and two members.

Execution Steps

1. Log in as manager.
2. Open channel management.
3. Create the project channel.
4. Add the specified members.
5. Save.
6. Open the channel as a member.
7. Attempt access as a non-member.

Expected Result: Channel and membership are saved; authorized members can access it; unauthorized user is denied.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-005 — Edit and delete own message

Field	Details
Test Type	Functional
Priority	Medium
Preconditions	Employee has an existing message in an editable conversation.
Test Data	Existing test message.

Execution Steps

1. Send a test message.
2. Use the message edit action.
3. Change its text and save.
4. Verify updated content.
5. Delete the same message.
6. Refresh the conversation.

Expected Result: Only permitted actions are available; edited content persists; deleted content follows the configured deletion policy and is no longer presented as active message content.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-006 — Search accessible message history

Field	Details
Test Type	Functional
Priority	Medium
Preconditions	Accessible messages contain a unique keyword.
Test Data	Unique keyword and optional date/channel filter.

Execution Steps

1. Open search.
2. Enter the unique keyword.
3. Run the search.
4. Apply a supported filter.
5. Open a returned result.

Expected Result: Relevant accessible results are returned; inaccessible conversations are excluded; filters work and the selected result opens correct context.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-007 — Mention and notification delivery

Field	Details
Test Type	Functional

Priority	Medium
Preconditions	Two users share a channel and notifications are enabled.
Test Data	Mention of Employee B.

Execution Steps

1. As Employee A, open shared channel.
2. Compose a message mentioning Employee B.
3. Send it.
4. Check Employee B notification center.
5. Open the notification.

Expected Result: Employee B receives one appropriate notification and the notification links to the correct channel/message.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-008 — Announcement targeting

Field	Details
Test Type	Functional
Priority	High
Preconditions	Manager/admin can publish targeted announcements; two users represent different audiences.
Test Data	Department-targeted announcement.

Execution Steps

1. Create an announcement targeted to Department A.
2. Publish it.
3. Log in as a Department A user.
4. Verify visibility.
5. Log in as a Department B user.
6. Verify visibility.

Expected Result: Targeted audience sees the announcement; non-targeted user cannot access restricted announcement content.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-009 — Valid attachment upload/download

Field	Details
Test Type	Functional
Priority	High
Preconditions	User is in an authorized conversation; permitted

	file type is configured.
Test Data	Small permitted PDF/image/document within limit.

Execution Steps

1. Open authorized conversation.
2. Select permitted test file.
3. Upload and send.
4. Wait for completion.
5. Download the attachment.
6. Verify file opens correctly.

Expected Result: Upload succeeds, message references the attachment, authorized recipient can download it, and file content remains intact.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-010 — Invalid and oversized attachment rejection

Field	Details
Test Type	Security
Priority	Critical
Preconditions	Upload validation is enabled.
Test Data	Unsupported extension; file exceeding configured size; safe malformed test file.

Execution Steps

1. Open authorized conversation.
2. Attempt upload of unsupported file.
3. Attempt upload of oversized file.
4. Attempt approved security-test sample.
5. Observe responses and storage.

Expected Result: Disallowed uploads are rejected, no executable content is exposed, clear validation messages are shown, and rejected files are not stored as accessible attachments.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-011 — Administrator user lifecycle

Field	Details
Test Type	Functional
Priority	High
Preconditions	Administrator account and test user are available.
Test Data	Synthetic user profile.

Execution Steps

1. Log in as administrator.
2. Create or activate a test user.
3. Assign standard role.
4. Verify account appears in user list.
5. Deactivate the account.
6. Attempt login using that account.

Expected Result: Admin changes persist and deactivated user cannot authenticate or access protected application functions.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-012 — Role-based authorization

Field	Details
Test Type	Security
Priority	Critical
Preconditions	Standard employee and administrator accounts exist.
Test Data	Administrator-only function/API endpoint.

Execution Steps

1. Log in as standard employee.
2. Attempt administrator-only UI action.
3. If authorized test tooling is available, attempt the corresponding protected API action.
4. Log in as administrator.
5. Repeat the action.

Expected Result: Standard employee receives access denied and no privileged operation occurs; administrator can perform the authorized action.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-013 — Session timeout and protected resources

Field	Details
Test Type	Security
Priority	High
Preconditions	Session timeout policy is configured.
Test Data	Active test account.

Execution Steps

1. Log in.
2. Leave the session inactive beyond configured timeout.
3. Attempt to open protected page.

4. Attempt a protected action/API call.

5. Observe authentication state.

Expected Result: Expired session is rejected; protected data is not returned; user must authenticate again.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-014 — Input validation and output encoding

Field	Details
Test Type	Security
Priority	Critical
Preconditions	Message and search fields are available.
Test Data	Boundary-length text; HTML/script-like test strings; special characters.

Execution Steps

1. Submit minimum-length valid input.
2. Submit maximum allowed length.
3. Submit over-limit input.
4. Submit HTML/script-like characters.
5. Open the resulting content.

Expected Result: Valid boundary input works; invalid input is rejected or safely handled; no script executes; application remains stable.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-015 — Audit logging of privileged actions

Field	Details
Test Type	Security/Audit
Priority	High
Preconditions	Admin can perform a controlled account/role change; audit viewer is available.
Test Data	Synthetic user and role change.

Execution Steps

1. Log in as administrator.
2. Change test user's role or account status.
3. Open audit log with authorized audit account.
4. Locate the event.
5. Compare event details with the performed action.

Expected Result: Audit record contains actor, action, target, timestamp and outcome, without exposing unnecessary secrets.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-016 — Concurrent-user messaging performance

Field	Details
Test Type	Performance
Priority	High
Preconditions	Performance environment is isolated and monitored; 100 synthetic users prepared.
Test Data	100 concurrent users; representative message payloads.

Execution Steps

1. Start monitoring CPU, memory, database and error metrics.
2. Ramp to 100 concurrent users.
3. Have users send representative messages for the agreed duration.
4. Record response times, throughput and errors.
5. Stop load and inspect logs/data integrity.

Expected Result: System remains stable; target response time is approximately ≤ 2 seconds for normal requests; error rate target is $< 1\%$; no message loss, duplication or corruption occurs.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-017 — Recovery after temporary service interruption

Field	Details
Test Type	Reliability
Priority	High
Preconditions	Recovery scenario is approved in staging and monitoring is active.
Test Data	Synthetic conversation/message.

Execution Steps

1. Send and confirm a test message.
2. Simulate a controlled temporary application dependency interruption.
3. Observe user-facing behavior.
4. Restore the dependency.
5. Retry the operation if required.
6. Verify message/data consistency.

Expected Result: System fails gracefully, recovers after dependency restoration, does not create duplicate/lost messages, and records useful error/monitoring information.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-018 — Responsive browser usability

Field	Details
Test Type	Usability
Priority	Medium
Preconditions	Supported browsers/devices are available.
Test Data	Standard employee account.

Execution Steps

1. Log in using Chrome desktop.
2. Complete send-message workflow.
3. Repeat using Edge.
4. Resize to a mobile viewport or use supported mobile browser.
5. Complete navigation and message workflow.

Expected Result: Controls remain usable, text is readable, key actions are discoverable, and no critical layout or navigation issue blocks common tasks.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-019 — Regression after defect correction

Field	Details
Test Type	Regression
Priority	High
Preconditions	A previously verified defect has been fixed in a new build.
Test Data	Defect reproduction data and affected workflow.

Execution Steps

1. Review the defect's original reproduction steps.
2. Execute the original steps on the fixed build.
3. Run directly related functional cases.
4. Run the smoke/regression suite for impacted components.
5. Compare results with the previous baseline.

Expected Result: Original defect no longer reproduces and related/core functionality continues to pass without new Critical/High defects.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

TC-020 — Database/search consistency after repeated operations

Field	Details
Test Type	Integration
Priority	High
Preconditions	Staging database and search service are available.
Test Data	Synthetic messages with unique IDs/keywords.

Execution Steps

1. Create several messages in a test channel.
2. Edit permitted messages.
3. Delete one according to policy.
4. Search using unique keywords.
5. Refresh and repeat the search.
6. Compare visible results with the expected final state.

Expected Result: Search and persisted data agree with the final message state; edited/deleted records follow policy; no stale or unauthorized result is returned.

Pass/Fail Criteria: PASS only if every expected result occurs. Otherwise mark FAIL and raise a defect with evidence.

12. Defect and Issue Management

A failed test is not automatically a software defect; the tester first confirms the environment, test data, build and reproduction steps. Confirmed defects are recorded with sufficient evidence for another team member to reproduce.

Field	Required Information
Identification	Defect ID, title, reporter and date
Context	Build, environment, browser/device and related requirement/test case
Reproduction	Preconditions, exact steps, test data and evidence
Expected vs Actual	What should happen versus what happened
Classification	Severity and priority
Ownership	Assigned developer/team and target build
Lifecycle	New → Triaged → Assigned → In Progress → Fixed → Retest → Closed/Reopened
Verification	Retest result and regression impact

12.1 Severity Guidance

- Critical — security bypass, data loss, privilege escalation, complete service failure or release-blocking issue.
- High — major business function unavailable or materially incorrect; serious but contained security/reliability issue.

- Medium — limited functional issue with a workaround.
- Low — cosmetic or minor usability issue with negligible business impact.

12.2 Bug Handling Process

19. Capture the failure with exact reproduction steps and evidence.
20. Confirm the failure and rule out environment/test-data problems.
21. Create and triage the defect with severity and priority.
22. Assign the defect to the responsible developer/team.
23. Developer fixes the issue and provides the corrected build.
24. QA retests the original case.
25. Run impacted regression cases.
26. Close the defect if verified; otherwise reopen it with new evidence.

13. Risk Management

Risk	Impact	Mitigation
Unstable staging environment	High	Smoke-test deployments; record environment blockers separately.
Insufficient test data	Medium	Prepare synthetic accounts/data before execution.
Third-party integration unavailable	High	Use approved mocks or schedule dependency testing.
Performance test affects other testing	High	Isolate load environment and schedule performance run.
Security defect discovered late	Critical	Run security tests early and gate release on critical findings.
Incomplete regression after fixes	High	Map every defect to impacted requirements and rerun related cases.
Test evidence missing	Medium	Require screenshots/log references and execution records for failures.

14. Performance and Security Acceptance Criteria

14.1 Performance

- Target approximately ≤ 2 seconds for common requests at baseline load.
- Support at least 100 concurrent synthetic users for the planned performance run.
- Target normal-operation error rate below 1%.
- Monitor CPU, memory, database utilization, response-time percentiles and throughput.
- No message loss, duplication or corruption is acceptable.

14.2 Security

- Unauthorized users must not access restricted functions or data.
- Authentication failures must not disclose sensitive account information.

- Expired sessions must not access protected resources.
- User input must be safely validated/encoded.
- Disallowed or unsafe uploads must be rejected or safely isolated.
- Privileged actions must be auditable.
- Critical security defects block release unless formally accepted by authorized management.

15. Requirements Traceability

Requirement	Test Cases	Coverage
FR-01 Authentication	TC-001, TC-002	Functional + Security
FR-02 Messaging	TC-003, TC-005, TC-016, TC-020	Functional + Performance + Integration
FR-03 Channels	TC-004	Functional
FR-04 Search	TC-006, TC-020	Functional + Integration
FR-05 Notifications	TC-007	Functional
FR-06 Announcements	TC-008	Functional
FR-07 File Sharing	TC-009, TC-010	Functional + Security
FR-08 Administration	TC-011, TC-015	Functional + Audit
FR-09 Access Control	TC-012	Security
NFR-01 Performance	TC-016	Performance
NFR-02 Session Security	TC-013	Security
NFR-03 Input/File Security	TC-010, TC-014	Security
NFR-04 Availability/Recovery	TC-017	Reliability
NFR-05 Usability	TC-018	Usability
NFR-06 Regression Stability	TC-019	Regression

16. Test Reporting and Metrics

- Execution: planned, executed, passed, failed, blocked and not-run cases.
- Coverage: percentage of approved requirements mapped to at least one test.
- Defects: total defects by severity/priority, open/closed count, reopen rate and average resolution time.
- Performance: average and percentile response times, throughput, concurrent users and error rate.
- Security: findings by severity and remediation status.
- Release readiness: unresolved blockers, accepted risks and final recommendation.

17. Post-Testing Review

27. Confirm all planned tests have a recorded status.
28. Review Critical and High defects with stakeholders.
29. Verify all accepted deviations have documented owners and rationale.
30. Summarize functional, performance and security results.
31. Identify lessons learned and improvements for the next test cycle.
32. Archive evidence, execution records and the final test summary.
33. Provide a release recommendation: Ready, Ready with Accepted Risks, or Not Ready.

18. Conclusion

This Week 3 Test Plan provides a comprehensive and practical approach for verifying the Internal Communication Platform against its approved requirements. It combines structured functional testing with integration, regression, usability, performance, security and reliability checks. The twenty detailed cases cover the highest-value employee and administrator workflows and address major risks such as unauthorized access, unsafe uploads, session expiry, input handling, concurrent usage and data consistency.

The defined environment, synthetic test data, schedule, entry and exit criteria, defect lifecycle, risk controls, performance targets and requirements traceability provide a clear basis for execution and release decision-making. The document therefore demonstrates the analytical and quality-assurance responsibilities expected of a Junior Systems Analyst.